

Customer Revenue, Risk & Retention Analytics Project

Project Objective

This project focuses on analyzing customer behavior, revenue trends, returns, and delivery performance using Python and Pandas.

The goal is to generate business insights and support data-driven decision-making.

Data Preparation

The dataset was initially provided in Excel format with three sheets: customers, orders, and order items. Each sheet was converted into separate CSV files and then loaded into Python using Pandas.

Data Processing

All datasets were merged using `customer_id` and `order_id` to create a single dataset.

New columns were created such as `order_value`, `delivery_delay_days`, `is_repeat_customer`, `is_high_value_order`, `customer_value_segment`, `customer_risk_category`, and `order_reliability_score`.

Analysis Approach

The project was completed using Python with functions, lambda expressions, and for-loops.

Pandas was used for data manipulation and analysis.

Business Questions & Answers

1. Which customers generate high revenue but have high return rates?

Some customers generate high revenue but also return products frequently, making them risky for the business.

2. Which cities suffer most from delivery delays and revenue loss?

Cities like Mumbai and Chennai show higher delivery delays, indicating possible logistics issues.

3. Are repeat customers actually more valuable?

Repeat customers have a higher average order value, making them more valuable to the business.

4. Which categories appear profitable but lose revenue due to returns?

Categories like Clothing and Electronics generate high revenue but also have high return rates.

5. Which payment mode carries the highest risk?

UPI has the highest number of returns and cancellations, making it the riskiest payment mode.

Visualizations

The following charts were created:

- Monthly revenue trend (Line chart)
- Customer value segmentation (Bar chart)
- Return rate by category (Bar chart)
- Delivery delay vs revenue (Scatter plot)
- City vs category revenue (Heatmap)

Business Problems

1. High-value customers are also returning products frequently.
 2. Certain cities have higher delivery delays.
 3. Some product categories have high return rates.
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Recommendations

1. Improve product quality and provide accurate descriptions to reduce returns.
 2. Optimize delivery operations in high-delay cities.
 3. Focus on reducing returns in high-risk categories like Clothing and Electronics.
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Conclusion

This project demonstrates how Python and data analysis techniques can be used to understand customer behavior, identify risks, and improve business performance through data-driven insights.